AWS CloudWatch Hands-on Lab Tasks (Tasks 1–12)

Task 1: Explore the CloudWatch Console

Navigate through Metrics, Logs, Alarms, Dashboards, Events (EventBridge), and Application Signals. Identify AWS services publishing metrics.

Task 2: Monitor an EC2 Instance

Monitor CPU Utilization, Network In/Out, Disk Read/Write Bytes, and Status Check metrics for an EC2 instance.

Task 3: Install the CloudWatch Agent

Install and configure the CloudWatch Agent to collect memory, disk, and swap utilization metrics from an EC2 instance.

Task 4: Create a CloudWatch Dashboard

Build a dashboard displaying CPU, memory, disk usage, and network metrics for your EC2 instance.

Task 5: Create a CloudWatch Alarm

Create an alarm that triggers when CPU utilization exceeds 70% and configure an SNS notification.

Task 6: Generate High CPU Utilization

Generate CPU load using a stress tool and observe the alarm transition from OK to ALARM.

Task 7: Configure Amazon SNS Notifications

Create an SNS topic, subscribe an email address, and integrate it with your CloudWatch Alarm.

Task 8: Monitor an Amazon RDS Database

Monitor RDS metrics such as CPU Utilization, Database Connections, Free Storage Space, Read IOPS, and Write IOPS.

Task 9: Explore CloudWatch Logs

Create a Log Group and configure the CloudWatch Agent to send system logs to CloudWatch Logs.

Task 10: Create a Metric Filter

Create a Metric Filter to count ERROR or WARNING log entries and generate a custom metric.

Task 11: Create an Alarm Using Log Metrics

Create an alarm based on the custom metric and send notifications through SNS when the threshold is exceeded.

Task 12: Create an AWS Billing Alarm

Enable billing alerts and create an alarm to notify you when estimated AWS charges exceed a specified amount.